



BAULKHAM HILLS HIGH SCHOOL

2011
YEAR 12 HALF YEARLY
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided on the back of the cover sheet
- All necessary working should be shown in every question
- Marks may be deducted for careless or badly arranged work

Total marks – 80

- Attempt Questions 1 – 8
- All questions are of equal value
- Start a separate piece of paper for each question.
- Put your BOS number and the question number at the top of each sheet.

Total marks – 80

Attempt Questions 1 – 8

All questions are of equal value

Answer each question on a SEPARATE piece of paper clearly marked Question 1, Question 2, etc. Each piece of paper must show your BOS number.

	Marks
Question 1 (10 marks) Use a <i>separate</i> piece of paper	
a) Given that $z = -\sqrt{3} - i$, evaluate;	
(i) $\bar{z}$	1
(ii) $ z $	1
(iii) $\arg z$	1
(iv) $\frac{1}{z}$	1
(v) z^4 in the form $x + iy$	2
b) (i) On an Argand Diagram sketch the locus of a point corresponding to the complex number z , where $ z - 3 = 3$	2
(ii) Use your diagram in (i) to explain why $\arg(z - 3) = \arg z^2$	2
Question 2 (10 marks) Use a <i>separate</i> piece of paper	
a) For the hyperbola $\frac{y^2}{12} - \frac{x^2}{4} = 1$, find the;	
(i) eccentricity.	1
(ii) coordinates of the foci.	1
(iii) equations of the directrices.	1
(iv) equations of the asymptotes.	1
b) The polynomial $P(z)$ has the equation $P(z) = z^4 - 2z^3 - z^2 + 2z + 10$. Given that $(z - 2 + i)$ is a factor of $P(z)$, express $P(z)$ as a product of two real quadratic factors.	3
c) Solve the inequality $\frac{4x}{3 + 2x - x^2} \leq 1$	3

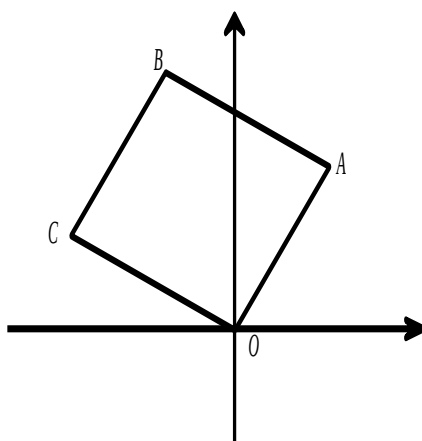
Marks

Question 3 (10 marks) Use a *separate* piece of paper

- a) (i) Find a monic cubic equation with roots α, β and γ such that $\alpha\beta\gamma = 5$, $\alpha + \beta + \gamma = 7$ and $\alpha^2 + \beta^2 + \gamma^2 = 29$. 2
- (ii) Find $\alpha^3 + \beta^3 + \gamma^3$ 2
- b) (i) Given that $z = \cos \theta + i \sin \theta$, prove that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ 2
- (ii) Hence, or otherwise, solve $2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$ 4

Question 4 (10 marks) Use a *separate* piece of paper

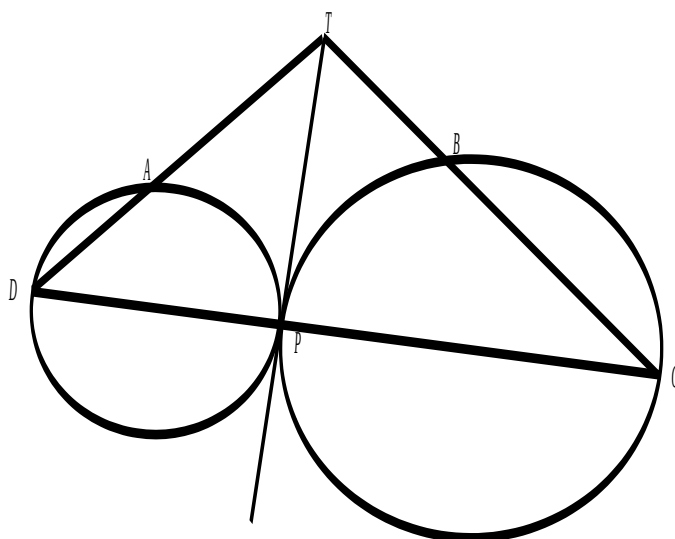
- a) In the square $OABC$, shown below, the point A represents the complex number z and the point C represents the complex number ω .



- (i) Find $\arg\left(\frac{z - \omega}{z + \omega}\right)$, 2
- (ii) If $z = 2 + i$, what complex numbers do B and C represent? 3
- b) How many different ways are there of arranging four married couples at a circular table with men and women in alternate positions, and no wife next to her husband? 2
- c) Prove that $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} = 0$ 3
has no multiple roots for $n \geq 1$

Question 5 (10 marks) Use a *separate* piece of paper

- a) The equation $x^3 - 6x^2 + 11x - 6 = 0$, has roots α, β and γ .
- (i) Without finding the roots α, β and γ , find the equation with roots $\alpha - 2, \beta - 2$ and $\gamma - 2$. 2
- (ii) Hence, or otherwise, solve $x^3 - 6x^2 + 11x - 6 = 0$. 3
- b) In the diagram, two circles touch at P and share a common tangent TP .
 C lies on the circumference of one circle and D lies on the circumference of the other, such that D, P and C are collinear.
 TC and TD intersect the circles at B and A respectively.



(Note: on page 7 there is a copy of the above diagram that you can use, please detach and include with your solutions)

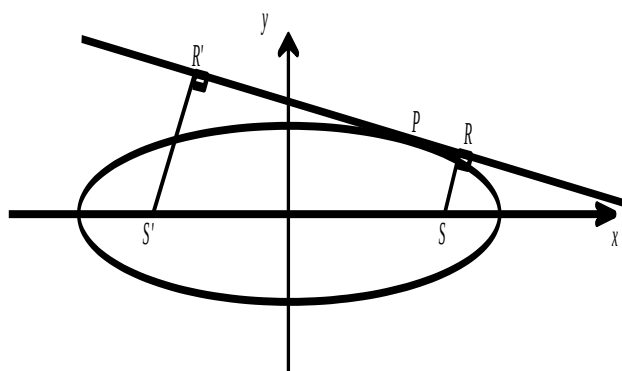
- (i) Show, giving reasons, that $ATBP$ is a cyclic quadrilateral. 3
- (ii) Show, giving reasons, that $ABCD$ is a cyclic quadrilateral. 2

Question 6 (10 marks) Use a *separate* piece of paper

- a) Let ω be one of the cube roots of unity.
- (i) Find the two possible values of $1 + \omega + \omega^2$. 2
- (ii) If ω is one of the non-real cube roots of unity, find the value of $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$ 2
- b) The tangent to the rectangular hyperbola $xy = 4$ at the point $T\left(2t, \frac{2}{t}\right)$ has the equation $x + t^2y = 4t$. (Do not prove this)
- (i) The tangent cuts the x -axis at the point Q . Show that the line through Q which is perpendicular to the tangent at T has equation $t^2x - y = 4t^3$. 2
- (ii) This line through Q cuts the rectangular hyperbola at the points R and S . Show that the midpoint M of RS has coordinates $M(2t, -2t^3)$. 2
- (iii) Find the equation of the locus of M as T moves on the rectangular hyperbola, stating any restrictions that may apply. 2

Question 7 (10 marks) Use a *separate* piece of paper

- a) The complex number $\frac{\sqrt{3}}{2} + \frac{i}{2}$ is one of the n th roots of -1 . 2
Find the least value of n for this to be so.
- b) Solve the equation $x^4 + x^3 + x^2 + x + 1 = 0$ 3
- c) Consider the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where $a > b > 0$.



- (i) Show that the tangent to the ellipse at the point $(a \cos \theta, b \sin \theta)$ has the equation $bx \cos \theta + ay \sin \theta - ab = 0$. 2
- (ii) R and R' are the feet of the perpendiculars from the foci S and S' to the tangent at P . Show that $SR \times S'R' = b^2$. 3

Question 8 (10 marks) Use a *separate* piece of paper

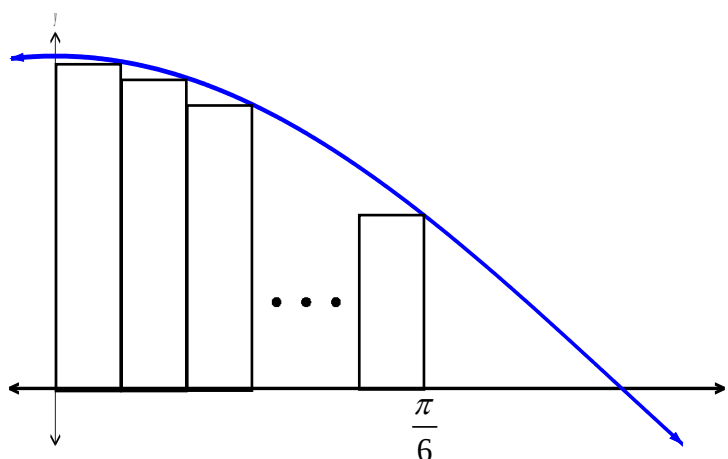
- a) 47231 is a five digit number whose digits sum to 17 (i.e. $4+7+2+3+1=17$)
How many five digit numbers sum to 43? 2

- b) (i) Prove by mathematical induction, that for all positive integers n ; 4

$$\frac{1}{2} + \cos \theta + \cos 2\theta + \dots + \cos (n-1)\theta = \frac{\sin \frac{1}{2}(2n-1)\theta}{2 \sin \frac{1}{2}\theta}$$

(Hint: you may use the identity $2 \cos A \sin B = \sin(A+B) - \sin(A-B)$)

- (ii) In the diagram, n rectangles of equal width are constructed under the curve 2
 $y = \cos x$ from $x = 0$ to $x = \frac{\pi}{6}$.



Use the result of part (i) to show that the sum S_n of the areas of the n rectangles is given by

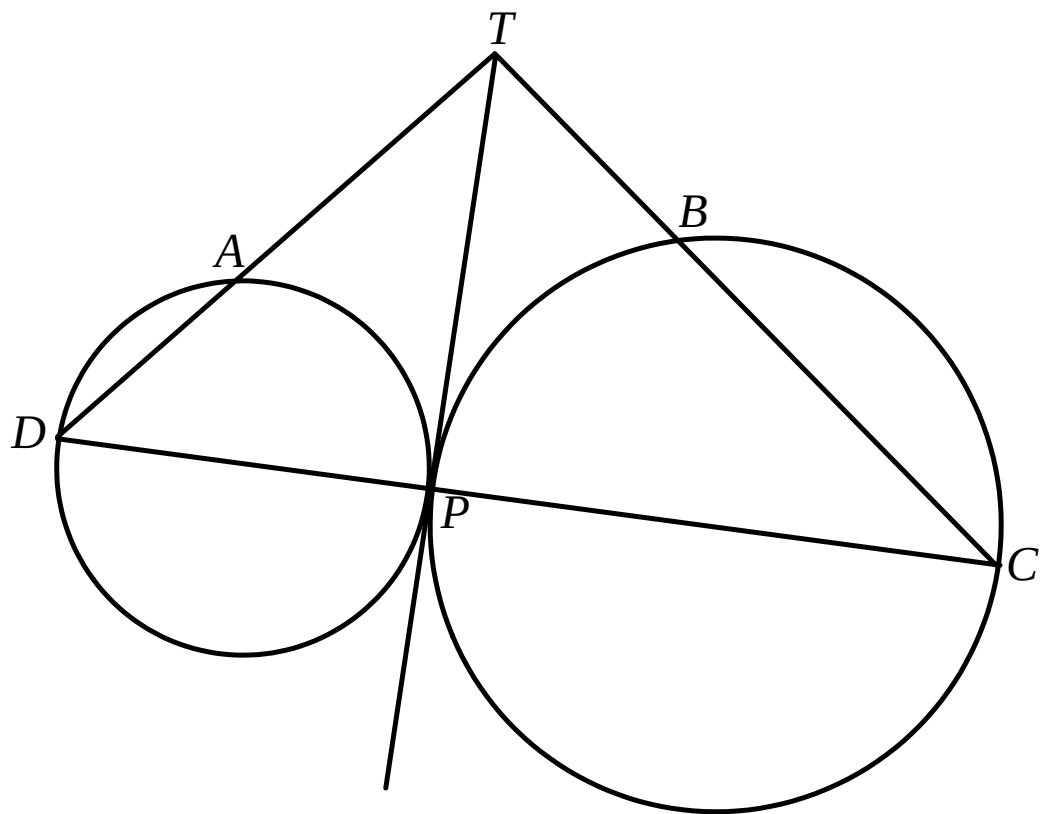
$$S_n = \frac{\pi}{12n} \left\{ (\sqrt{3}-1) + \frac{\sin(2n-1)\frac{\pi}{12n}}{\sin \frac{\pi}{12n}} \right\}$$

- (iii) Hence find $\lim_{n \rightarrow \infty} S_n$ 2
(Hint: Let $n = \frac{1}{h}$)

End of paper

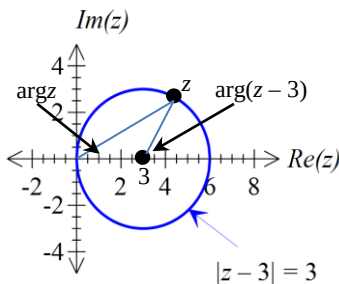
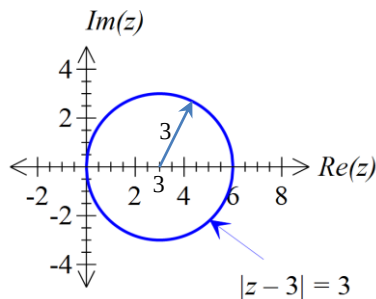
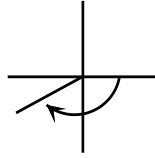
Question 5 b)

Please detach and include with your solutions



BAULKHAM HILLS HIGH SCHOOL
EXTENSION 2 HALF YEARLY 2011 SOLUTIONS

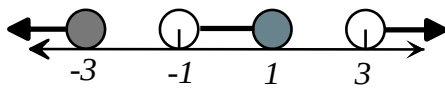
Solution	Marks	Comments
QUESTION 1		
1a) (i) $\bar{z} = -\sqrt{3} + i$	1	
1a) (ii) $ z = \sqrt{(-\sqrt{3})^2 + (-1)^2}$ $= 2$	1	
1a) (iii) $\arg z = \tan^{-1}\left(\frac{-1}{-\sqrt{3}}\right)$ $= -\frac{5\pi}{6}$	1	<i>Note: argument should be quoted $-\pi < \arg z \leq \pi$</i>
1a) (iv) $\frac{1}{z} = \frac{\bar{z}}{ z ^2}$ $= \frac{-\sqrt{3} + i}{4}$ $= -\frac{\sqrt{3}}{4} + \frac{i}{4}$	1	
1a) (v) $z^4 = \left(2\text{cis } -\frac{5\pi}{6}\right)^4$ $= 2^4 \text{cis } -\frac{20\pi}{6}$ $= 16\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ $= -8 + 8\sqrt{3}i$	2	1 mark • Correct use of De Moivre's theorem or equivalent merit • Correctly changing from mod-arg to $x + iy$ <i>Note: correct use of student's answers to (ii) & (iii) should gain full marks</i>
1b) (i)	2	1 mark • Recognises the locus is a circle
1b) (ii)	2	1 mark • Uses diagram to make progress towards a solution

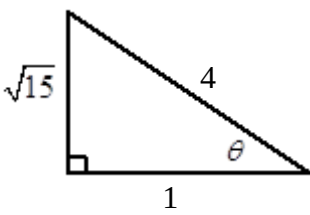
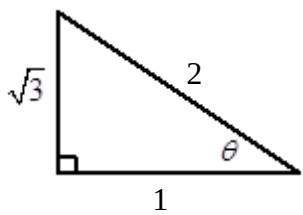


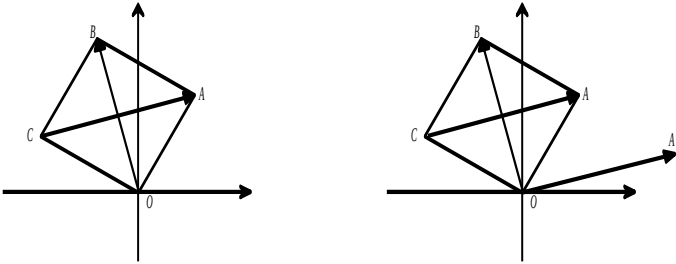
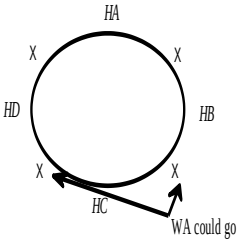
$$\arg(z - 3) = 2\arg z$$

$$\left(\angle \text{ at centre twice } \angle \text{ at circumference } \right)$$

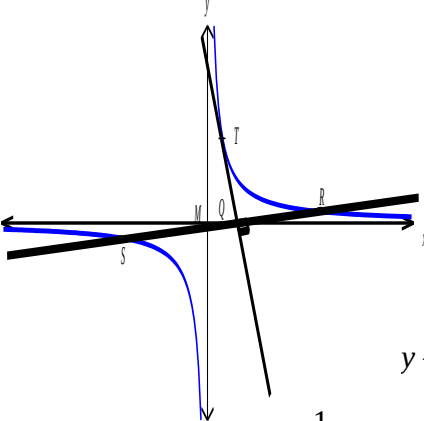
$$\therefore \arg(z - 3) = \arg z^2$$

Solution	Marks	Comments
QUESTION 2		
2a) (i) $a^2 = b^2(e^2 - 1)$ $4 = 12(e^2 - 1)$ $e^2 - 1 = \frac{1}{3}$ $e^2 = \frac{4}{3}$ $e = \frac{2}{\sqrt{3}}$	1	
2a) (ii) $\left(0, \pm\sqrt{12} \times \frac{2}{\sqrt{3}}\right)$ $= (0, \pm 4)$	1	1 mark • Correct use of formulae and foci located on x axis and directrices parallel to y axis
2a) (iii) $y = \pm\sqrt{12} \times \frac{\sqrt{3}}{2}$ $y = \pm 3$	1	
2a) (iv) $y = \pm\frac{\sqrt{12}}{2}x$ $y = \pm\sqrt{3}x$	1	
2b) If $(z - 2 + i)$ is a factor, so is $(z - 2 - i)$ $\therefore (z^2 - 4z + 5)$ is a factor $P(z) = z^4 - 2z^3 - z^2 + 2z + 10$ $= (z^2 - 4z + 5)(z^2 + 2z + 2)$	3	2 marks • Finds one quadratic factor 1 mark • Identifies conjugate as another root
2c) $\frac{4x}{3 + 2x - x^2} \leq 1$ $3 + 2x - x^2 \neq 0$ $x^2 - 2x - 3 \neq 0$ $(x - 3)(x + 1) \neq 0$ $x \neq 3 \text{ or } x \neq -1$ $\frac{4x}{3 + 2x - x^2} = 1$ $4x = 3 + 2x - x^2$ $x^2 + 2x - 3 = 0$ $(x + 3)(x - 1) = 0$ $x = -3 \text{ or } x = 1$  $x \leq -3 \text{ or } -1 < x \leq 1 \text{ or } x > 3$	3	3 marks • Correct graphical solution on number line or algebraic solution, with correct working 2 marks • Bald answer • Identifies the four correct critical points via a correct method • Correct conclusion to their critical points obtained using a correct method 1 mark • Uses a correct method • Acknowledges a problem with the denominator. 0 marks • Solves like a normal equation, with no consideration of the denominator.

Solution	Marks	Comments
QUESTION 3		
3a) (i) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $29 = 7^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma)$ $2(\alpha\beta + \alpha\gamma + \beta\gamma) = 20$ $\alpha\beta + \alpha\gamma + \beta\gamma = 10$ $\therefore$ equation is $x^3 - 7x^2 + 10x - 5 = 0$	2	1 mark - Correctly evaluates $\Sigma\alpha\beta$ - Correct equation for their $\Sigma\alpha\beta$
3a) (ii) $\sum \alpha^3 - 7 \sum \alpha^2 + 10 \sum \alpha - 15 = 0$ $\sum \alpha^3 = 7 \sum \alpha^2 - 10 \sum \alpha + 15$ $= 7(29) - 10(7) + 15$ $= 148$	2	2 marks - Correctly solution for incorrect equation found in part (i) 1 mark - Correctly deriving a formula for $\Sigma\alpha^3$ using their equation. - Substitutes correctly into an incorrect $\Sigma\alpha^3$ obtained using a correct method.
3b) (i) $z^n + \frac{1}{z^n} = (\text{cis}\theta)^n + (\text{cis}\theta)^{-n}$ $= \text{cis}(n\theta) + \text{cis}(-n\theta)$ $= \cos n\theta + i \sin n\theta + \cos n\theta + i \sin(-n\theta)$ $= \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta$ $= 2 \cos n\theta$	2	1 mark Uses De Moivre's Theorem in an attempt to show desired result.
3b) (ii) $2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$ $2z^2 + 3z + 5 + \frac{3}{z} + \frac{2}{z^2} = 0$ $2\left(z^2 + \frac{1}{z^2}\right) + 3\left(z + \frac{1}{z}\right) + 5 = 0$ $4\cos 2\theta + 6\cos\theta + 5 = 0$ $8\cos^2\theta - 4 + 6\cos\theta + 5 = 0$ $8\cos^2\theta + 6\cos\theta + 1 = 0$ $(4\cos\theta + 1)(2\cos\theta + 1) = 0$ $\cos\theta = -\frac{1}{4} \quad \text{or} \quad \cos\theta = -\frac{1}{2}$   $\therefore z = -\frac{1}{4} \pm \frac{\sqrt{15}}{4}i \quad \text{or} \quad z = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$	4	3 marks - Correctly finding values for $\cos\theta$ - Correct solution for incorrect values of $\cos\theta$ 2 marks - Correctly finding the equation $4\cos 2\theta + 6\cos\theta + 5 = 0$ or equivalent 1 mark - Dividing equation by z^2

Solution	Marks	Comments
QUESTION 4		
4a) (i)  $\vec{OB} = z + \omega$ $\vec{CA} = z - \omega$ translate $\vec{CA}$ to origin $\angle BOA' = \frac{\pi}{2}$ (diagonals $\perp$ in square) $\arg\left(\frac{z - \omega}{z + \omega}\right) = \arg(z - \omega) - \arg(z + \omega)$ $= -[\arg(z + \omega) - \arg(z - \omega)]$ $= -\frac{\pi}{2}$	2	1 mark - Linking the problem to the diagonals of the square - Answer of $\frac{\pi}{2}$
4a) (ii) $C = iz$ $= i(2 + i)$ $= -1 + 2i$ $B = z + \omega$ $= 2 + i + (-1 + 2i)$ $= 1 + 3i$	3	2 marks - Correctly finds one of B or C and makes progress towards finding other. 1 mark - Correctly finds B or C - States $B = z + \omega$ and $C = iz$, but does not evaluate <i>Note: statements may be made in part (i)</i>
4b) ① Place husbands around table = 3! ② Wife A has two spots to choose ③ Wife B (or D) then only has one spot ④ Leaving only one spot for each of the other two wives. Ways = $3! \times 2 \times 1 \times 1 \times 1$ $= 12$ 	2	1 mark Partially correct logic that leads to an incorrect answer.
4c) $P(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$ $P'(x) = 1 + \frac{2x}{2!} + \frac{3x^2}{3!} + \dots + \frac{nx^{n-1}}{n!}$ $P'(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{n-1}}{(n-1)!}$ In order for multiple roots to exist $P(x) = P'(x) = 0$ $P(x) = P'(x)$ $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^{n-1}}{(n-1)!}$ $\frac{x^n}{n!} = 0$ $x = 0$ But $P(0) = 1 \neq 0$, $\therefore x = 0$ is not a root, thus no multiple roots exist.	3	2 marks - Finds $P'(x)$ and equates with $P(x)$ to find possible multiple root. 1 mark - Uses $P'(x) = 0$ to find multiple roots.

Solution	Marks	Comments
QUESTION 5		
5a) (i) Let $y = x - 2$ $x = y + 2$ $x^3 - 6x^2 + 11x - 6 = 0$ $(y + 2)^3 - 6(y + 2)^2 + 11(y + 2) - 6 = 0$ $y^3 + 6y^2 + 12y + 8 - 6y^2 - 24y - 24 + 11y + 22 - 6 = 0$ $y^3 - y = 0$	2	1 mark • Substitutes correctly for x .
5a) (ii) $y^3 - y = 0$ $y(y^2 - 1) = 0$ $y = 0$ or $y = \pm 1$ $\therefore x - 2 = 0$ or $x - 2 = \pm 1$ $x = 2$ or $x = 2 \pm 1$ $x = 2$ or $x = 1$ or $x = 3$	3	2 marks • Finds some of the correct roots of original equation • Finds all three roots using incorrect roots of equation found in (i) 1 mark • Finds roots of equation found in (i)
5b) (i) $\angle TPB = \angle BCP \quad (\text{alternate segment theorem})$ $\angle TPA = \angle ADP \quad (\text{ " " })$ $\angle TCD + \angle TDC + \angle CTD = 180^\circ \quad (\angle \text{ sum of } \triangle TCD)$ $\therefore \angle TPB + \angle TPA + \angle CTD = 180^\circ$ $\angle APB = \angle TPB + \angle TPA \quad (\text{common } \angle)$ $\angle APB + \angle CTD = 180^\circ$ $\therefore ATBP \text{ is a cyclic quadrilateral} \quad (\text{opposite } \angle \text{'s supplementary})$	3	2 marks • Correct solution with poor reasoning • Significant progress towards solution with good reasoning. 1 mark • Significant progress towards solution with poor reasoning. • Progress towards solution with good reasoning.
5b) (ii) $\angle TPA = \angle TBA \quad (\angle \text{'s in same segment are } =)$ $\therefore \angle PDA = \angle TBA \quad (\angle TPA = \angle PDA, \text{ proven in (i)})$ $\therefore ABCD \text{ is a cyclic quadrilateral} \quad (\text{exterior } \angle = \text{opposite interior } \angle)$	2	1 mark • Correct solution with poor reasoning • Progress towards solution with good reasoning.

Solution	Marks	Comments
QUESTION 6		
6a) (i) If ω is a complex cube root then other roots are 1 and ω^2 $\therefore 1 + \omega + \omega^2 = 0$ (sum of roots of $x^3 - 1 = 0$) If ω is a real cube root then $\omega = 1$ $\therefore 1 + \omega + \omega^2 = 1 + 1 + 1^2$ $= 3$ $\therefore 1 + \omega + \omega^2 = 0$ or 3	2	1 mark Correctly find one possible value.
6a) (ii) $(2 - \omega)(2 - \omega^2)(2 - \omega^4)(2 - \omega^5)$ ($\omega^3 = 1$) $= (2 - \omega)(2 - \omega^2)(2 - \omega)(2 - \omega^2)$ $= (2 - \omega)^2(3 + \omega)^2$ ($\omega^2 = -1 - \omega$) $= (6 - \omega - \omega^2)^2$ $= (7)^2$ $= 49$ ($1 = -\omega - \omega^2$)	2	1 mark - Uses $\omega^3 = 1$. - Uses $1 + \omega + \omega^2 = 0$
6b) (i)  x intercepts occur when $y = 0$ $t^2x - 0 = 4t^3$ $t^2x = 4t^3$ $x = 4t$ Slope of tangent $= -\frac{1}{t^2}$ $\therefore$ required slope $= t^2$ $y - 0 = t^2(x - 4t)$ $y = t^2x - 4t^3$ $t^2x - y - 4t^3 = 0$	2	1 mark - Finds x intercept. - Finds slope
6b) (ii) Solving line and hyperbola simultaneously $xy = 4$ $x(t^2x - 4t^3) = 4$ $t^2x^2 - 4t^3x - 4 = 0$ The roots of this equation are the x coordinates of R and S. $x_M = \frac{\Sigma\alpha\beta}{2}$ $= \frac{4t^3}{2t^2}$ $= 2t$ $y_M = t^2(2t) - 4t^3$ $= 2t^3 - 4t^3$ $= -2t^3$ $\therefore M(2t, -2t^3)$	2	1 mark - Successfully finds x or y coordinate - Uses $\Sigma\alpha\beta$ in attempt to find midpoint
6b) (iii) $x = 2t$ $y = -2t^3$ $t = \frac{x}{2}$ $= -2\left(\frac{x}{2}\right)^3$ $= -\frac{x^3}{4}$ However $t \neq 0$, $\therefore x \neq 0$ Locus is $y = -\frac{x^3}{4}$, excluding the point $(0,0)$	2	1 mark Identifies locus without the point of exclusion

Solution		Marks	Comments
QUESTION 7			
7a)	$\frac{\sqrt{3}}{2} + \frac{i}{2} = \text{cis} \frac{\pi}{6}$ $\left(\text{cis} \frac{\pi}{6} \right)^n = -1$ $\text{cis} \frac{n\pi}{6} = \text{cis} \pi$ $\frac{n\pi}{6} = \pi$ $\therefore n = 6$	2	1 mark • Another multiple of 6. • $\text{cis} \frac{n\pi}{6} = \text{cis} \pi$
7b)	$x^4 + x^3 + x^2 + x + 1 = 0$ $\frac{(x-1)(x^4 + x^3 + x^2 + x + 1)}{x-1} = 0$ $\frac{x^5 - 1}{x-1} = 0$ $x^5 - 1 = 0, \quad x \neq 1$ $x = \text{cis} \frac{2\pi}{5}, \text{cis} -\frac{2\pi}{5}, \text{cis} \frac{4\pi}{5}, \text{cis} -\frac{4\pi}{5}$	3	2 marks • Finds the solutions, but includes $x = 1$ • Incorrectly solves $x^5 - 1 = 0$ but acknowledges $x \neq 1$ 1 mark • Attempts to link the question with roots of unity
7c) (i)	$x = a \cos \theta \quad y = b \sin \theta \quad \frac{dy}{dx} = \frac{dy}{d\theta} \times \frac{d\theta}{dx}$ $\frac{dx}{d\theta} = -a \sin \theta \quad \frac{dy}{d\theta} = b \cos \theta$ $= -\frac{b \cos \theta}{a \sin \theta}$ $y - b \sin \theta = -\frac{b \cos \theta}{a \sin \theta} (x - a \cos \theta)$ $a y \sin \theta - a b \sin^2 \theta = -b x \cos \theta + a b \cos^2 \theta$ $b x \cos \theta + a y \sin \theta - a b (\sin^2 \theta + \cos^2 \theta) = 0$ $b x \cos \theta + a y \sin \theta - a b = 0$	2	1 mark • Correctly derives slope of the tangent
7c) (ii)	$SR = \left \frac{ab \cos \theta - 0 - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right \quad S'R' = \left \frac{-ab \cos \theta - 0 - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right $ $SR \times S'R' = \left \frac{ab \cos \theta - 0 - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right \times \left \frac{-ab \cos \theta - 0 - ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}} \right $ $= \frac{ a^2 b^2 e^2 \cos^2 \theta - a^2 b^2 }{b^2 \cos^2 \theta + a^2 \sin^2 \theta}$ $= b^2 \left(\frac{ a^2 e^2 \cos^2 \theta - a^2 }{b^2 \cos^2 \theta + a^2 \sin^2 \theta} \right)$ $= b^2 \left\{ \frac{ a^2 e^2 \cos^2 \theta - a^2 }{a^2 (1 - e^2) \cos^2 \theta + a^2 \sin^2 \theta} \right\}$ $= b^2 \left\{ \frac{ -1 a^2 - a^2 e^2 \cos^2 \theta }{a^2 (\sin^2 \theta + \cos^2 \theta) - a^2 e^2 \cos^2 \theta} \right\} = b^2 \left(\frac{a^2 - a^2 e^2 \cos^2 \theta}{a^2 - a^2 e^2 \cos^2 \theta} \right) = b^2$	3	2 marks • Makes substantial progress towards solution. 1 mark • Finds a correct expression for $SR \times S'R'$

Solution		Marks	Comments
QUESTION 8			
8a) Possible sets of digits $9 + 9 + 9 + 9 + 7 = 43$ $9 + 9 + 9 + 8 + 8 = 43$	# of numbers $= \frac{5!}{4!} + \frac{5!}{3! \times 2!}$ $= 5 + 10$ $= 15$	2	1 mark - Identifies the two sets of digits - Evaluates the number of ways for one set of digits
8b) (i) Prove for $n = 1$ $LHS = T_1$ $= \frac{1}{2}$ $\therefore LHS = RHS$ Hence the result is true for $n = 1$ Assume the result is true for $n = k$, where k is a positive integer i.e. $\frac{1}{2} + \cos\theta + \cos(2\theta) + \dots + \cos(k-1)\theta = \frac{\sin\frac{1}{2}(2k-1)\theta}{2\sin\frac{1}{2}\theta}$ Prove the result true for $n = k + 1$ i.e. Prove $\frac{1}{2} + \cos\theta + \cos(2\theta) + \dots + \cos k\theta = \frac{\sin\frac{1}{2}(2k+1)\theta}{2\sin\frac{1}{2}\theta}$ PROOF: $\frac{1}{2} + \cos\theta + \cos(2\theta) + \dots + \cos(k-1)\theta + \cos k\theta$ $= \frac{\sin\frac{1}{2}(2k-1)\theta}{2\sin\frac{1}{2}\theta} + \cos k\theta$ $= \frac{\sin\frac{1}{2}(2k-1)\theta + 2\sin\frac{1}{2}\theta \cos k\theta}{2\sin\frac{1}{2}\theta}$ $= \frac{\sin\frac{1}{2}(2k-1)\theta + \sin\left(k\theta + \frac{1}{2}\theta\right) - \sin\left(k\theta - \frac{1}{2}\theta\right)}{2\sin\frac{1}{2}\theta}$ $= \frac{\sin\frac{1}{2}(2k-1)\theta + \sin\frac{1}{2}(2k+1)\theta - \sin\frac{1}{2}(2k-1)\theta}{2\sin\frac{1}{2}\theta}$ $= \frac{\sin\frac{1}{2}(2k+1)\theta}{2\sin\frac{1}{2}\theta}$		4	There are 4 key parts of the induction; - Proving the result true for $n = 1$ - Clearly stating the assumption and what is to be proven - Using the assumption in the proof - Correctly proving the required statement 4 marks - Successfully does all of the 4 key parts 3 marks - Successfully does 3 of the 4 key parts 2 marks - Successfully does 2 of the 4 key parts 1 mark - Successfully does 1 of the 4 key parts

Solution	Marks	Comments
QUESTION 8...continued		
Hence the statement is true for $n = k + 1$, if it is true for $n = k$ Since the result is true for $n = 1$, then it is true for all positive integers, by induction.		
8b) (ii) The width of each rectangle is $\frac{\pi}{6n}$ $ \begin{aligned} S_n &= \frac{\pi}{6n} \left[\cos \frac{\pi}{6n} + \cos \frac{2\pi}{6n} + \cos \frac{3\pi}{6n} + \dots + \cos \frac{(n-1)\pi}{6n} + \cos \frac{n\pi}{6n} \right] \\ &= \frac{\pi}{6n} \left[\cos \frac{\pi}{6n} + \cos \frac{2\pi}{6n} + \cos \frac{3\pi}{6n} + \dots + \cos \frac{(n-1)\pi}{6n} + \cos \frac{\pi}{6} \right] \\ &= \frac{\pi}{6n} \left[\cos \frac{\pi}{6n} + \cos \frac{2\pi}{6n} + \cos \frac{3\pi}{6n} + \dots + \cos \frac{(n-1)\pi}{6n} + \frac{\sqrt{3}}{2} \right] \\ &= \frac{\pi}{6n} \left[\frac{\sqrt{3}}{2} - \frac{1}{2} + \left\{ \frac{1}{2} + \cos \frac{\pi}{6n} + \cos \frac{2\pi}{6n} + \cos \frac{3\pi}{6n} + \dots + \cos \frac{(n-1)\pi}{6n} \right\} \right] \\ &= \frac{\pi}{6n} \left\{ \frac{\sqrt{3}}{2} - \frac{1}{2} + \frac{\sin \frac{1}{2}(2n-1)\frac{\pi}{6n}}{2 \sin \frac{1}{2} \times \frac{\pi}{6n}} \right\} \\ &= \frac{\pi}{12n} \left\{ \sqrt{3} - 1 + \frac{\sin(2n-1)\frac{\pi}{12n}}{\sin \frac{\pi}{12n}} \right\} \end{aligned} $	2	1 mark • Successfully uses the result of part(i) in a method that would lead to a correct solution. • Writes S_n as a sum of the areas of the rectangles.
8b) (iii) $\lim_{n \rightarrow \infty} S_n = \lim_{h \rightarrow 0} S_{\frac{1}{h}}$ $ \begin{aligned} &= \lim_{h \rightarrow 0} \frac{h\pi}{12} \left[\sqrt{3} - 1 + \frac{\sin\left(\frac{2}{h} - 1\right)\frac{h\pi}{12}}{\sin \frac{h\pi}{12}} \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{(\sqrt{3} - 1)h\pi}{12} + \frac{h\pi}{12} \times \frac{\sin\left(\frac{2}{h} - 1\right)\frac{h\pi}{12}}{\sin \frac{h\pi}{12}} \right] \\ &= 0 + 1 \times \sin\left(\frac{\pi}{6} - 0\right) \\ &= \frac{1}{2} \end{aligned} $	2	1 mark • Bald answer • Uses $\lim_{x \rightarrow 0} \frac{x}{\sin x}$ • Correctly substitutes $n = \frac{1}{h}$